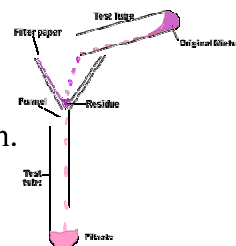


Separation Techniques

Filtration

Difference in particle size allows components to be separated using filter paper or any type of mesh or screen.
ex. wine making



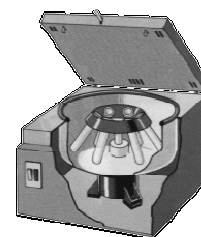
http://www.sciencebooks.ca/sci_content/science10/multimedia/17.html

Sedimentation

The effect of gravity on particles with differing densities in a liquid will cause suspended particles to settle in the bottom of the container.
ex. sewage treatment

Centrifugation

Much the same as sedimentation but this process is assisted by a machine (centrifuge) which speeds up the process
ex. cytology (study of cells and their components)



<http://glossary.pearsoned.com/dictionary.php?m=centrifuge>



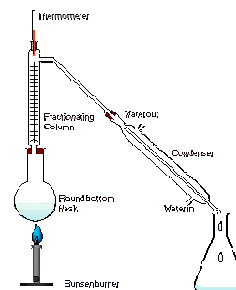
Magnetization

Some substances exhibit magnetic properties and therefore can be separated from non-magnetic substances using a magnet
ex. recycling

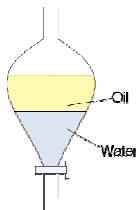
Distillation

Every pure substance has a specific boiling point. By boiling a mixture, the substance with the lowest b.p. can be caused to evaporate leaving behind the substance(s) with higher b.p.(s). The vapour from the boiling can be collected.

ex. crude oil processing



<http://www.canadaconnects.ca/chemistry/10104/>



<http://commons.wikimedia.org/wiki/File:SeparatoryFunnel.jpg>

Extraction

Liquids which are immiscible (solubility) will not mix to form a solution. A mixture of two or more immiscible liquids can be separated by allowing them to form layers according to density and then separating them with a separatory funnel.

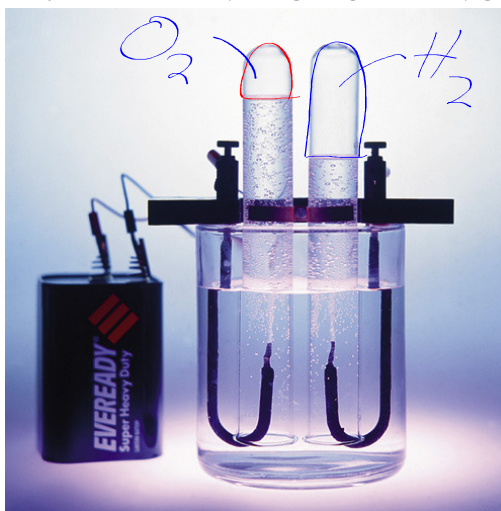
ex. organics

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Decomposition

By adding energy in the form of heat, electricity, light, etc. you can cause compounds to decompose into simpler compounds or elements. Note: elements do not decompose.

ex. decomposition of water into hydrogen gas & oxygen gas



<http://reich-chemistry.wikispaces.com/Decomposition+R.Stacey>

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